Amrita School of Artificial Intelligence, Faridabad
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Course Code: **23MAT214**

Mathematics for Intelligent System-4

Part- 2.3

Augmented Lagrangian and ADMM

BTech 4th Sem

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*All Sources: Stanford University, Texas Tech University, UC Santa Barbara, etc.
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Problem Setup (Motivation)

We consider constrained optimization:

$$\min_{x} f(x)$$

$$s.t.\ Ax = b$$

Direct handling is difficult because:

- constraints couple variables
- large-scale problems become expensive

So we move to a **Lagrangian formulation**.

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Classical Lagrangian

We introduce multiplier λ:

$$\mathcal{L}(x, \lambda) = f(x) + \lambda^T (Ax - b)$$

Idea:

- enforce constraint via dual variable
- solve saddle-point problem

⚠️ Issue:

- often unstable
- poor convergence numerically

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Augmented Lagrangian

The **Augmented Lagrangian** improves this by adding a quadratic penalty term:

$$\mathcal{L}_\rho(x, \lambda) = f(x) + \lambda^T(Ax - b) + \frac{\rho}{2} \| Ax - b \|_2^2$$

Here:

- $\rho > 0$ is a penalty parameter
- The quadratic term stabilizes convergence
- Combines **dual methods + penalty methods**
- $\lambda^T(Ax - b)$ →enforces constraint
- $\frac{\rho}{2} \| Ax - b \|^2$ →penalizes violation

👉 Intuition:

- Lagrangian → enforces constraint via multipliers
- Penalty term → forces feasibility strongly
- Together → faster and more robust optimization

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We start from the augmented Lagrangian:

$$\mathcal{L}_\rho(x, \lambda) = f(x) + \lambda^T(Ax - b) + \frac{\rho}{2} \| Ax - b \|_2^2$$

What we ideally want to do

$$x^{k+1} = \arg\min_x \mathcal{L}_\rho\left(x, \lambda^k\right)$$

$$\lambda^{k+1} = \lambda^k + \rho\left(Ax^{k+1} - b\right)$$

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❗ Core difficulty

The minimization step:

$$\min_x f(x) + \frac{\rho}{2} \| Ax - b \|^2 + \lambda^T (Ax - b)$$

👉 becomes **hard when**:

- $f(x)$ is complex (nonlinear, non-separable)
- A couples all variables
- high-dimensional problems (ML, imaging)

❗ Computational issue

Each iteration requires solving:

- a **large linear system** OR
- a **nonlinear optimization problem**

👉 This becomes a bottleneck in:

- large-scale ML
- image reconstruction
- distributed systems

❗ Structural issue: No separability

Suppose:

$$f(x) = f_1(x_1) + f_2(x_2)$$

But constraint:

$$Ax = b$$

👉 mixes x_1, x_2

So:

- cannot optimize each block independently
- no parallelization

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ADMM

(Alternating Direction Method of Multipliers).

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constrained optimization:

$$\min_{x} f(x)$$

$$s.t.\ Ax = b$$

Variable Splitting (Fundamental Step)

We rewrite the problem:

$$\min_{x,z} f(x) + g(z)\ s.t.\ Ax + Bz = c$$

Augmented Lagrangian with Splitting

Now define:

$$\mathcal{L}_\rho(x, z, \lambda) = f(x) + g(z) + \lambda^T(Ax + Bz - c) + \frac{\rho}{2} \| Ax + Bz - c \|^2$$

Why splitting works

- x handles smooth part
- z handles non-smooth / constraints
- constraint enforces consistency

👉 Now each variable can be optimized separately

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ADMM: Core Idea

We apply augmented Lagrangian:

$$\mathcal{L}_\rho(x, z, \lambda) = f(x) + g(z) + \lambda^T(Ax + Bz - c) + \frac{\rho}{2} \| Ax + Bz - c \|^2$$

Instead of minimizing jointly, we **alternate**.

ADMM Algorithm (Step-by-Step)

At iteration k:

Step 1: x-update

$$x^{k+1} = \arg\min_x \ \mathcal{L}_\rho\left(x, z^k, \lambda^k\right)$$

Step 2: z-update

$$z^{k+1} = \arg\min_z \ \mathcal{L}_\rho\left(x^{k+1}, z, \lambda^k\right)$$

Step 3: dual update

$$\lambda^{k+1} = \lambda^k + \rho\left(Ax^{k+1} + Bz^{k+1} - c\right)$$

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Key interpretation

- Split problem into **simpler subproblems**
- Solve them **alternately**
- Enforce consistency via dual updates

👉 That's why ADMM is powerful for:

- Large-scale optimization
- Distributed systems
- Machine learning problems

Notation and Variables"

- x: primal variable
- z: auxiliary variable
- u: scaled dual variable
- ρ: penalty parameter

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ADMM for Linear Programming (LP)

Standard LP form

$$\min_{x} \; c^T x \quad s.t.\, Ax = b,\; x \geq 0$$

We split:

- $x = z$
- One handles constraints, one handles positivity

Step 1: Reformulated:

$$\min_{x,z} \; c^T x + I_{z\geq 0}(z) \;\; s.t. \;\; x - z = 0$$

where $I_{z\geq 0}$ is an indicator function.

Step 2: Augmented Lagrangian

$$\mathcal{L}_\rho = c^T x + I_{z\geq 0}(z) + \frac{\rho}{2} \parallel x - z + u \parallel^2$$

$$I_{z\geq 0}(z) = \begin{cases} 0 & if\, z \geq 0 \\ +\infty & otherwise \end{cases}$$

- Indicator function is a **mathematical trick**
- Converts constraints into objective form
- Makes ADMM and proximal methods possible

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ADMM updates for LP

1. x-update

$$x^{k+1} = \arg\min_{x} \; c^T x + \frac{\rho}{2} \| x - z^k + u^k \|^2$$

This becomes a **quadratic problem with linear term**, often solved analytically.

2. z-update (projection step)

$$z^{k+1} = \Pi_{\mathbb{R}_+}\left(x^{k+1} + u^k\right)$$

👉 Simply **project onto non-negative orthant**:

$$z_i = \max(0, x_i)$$

3. .Dual update

$$u^{k+1} = u^k + \left(x^{k+1} - z^{k+1}\right)$$

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ADMM for Quadratic Programming (QP)

Standard QP $\min_x \ \frac{1}{2}x^T Qx + q^T x \ s.t. \ Ax = b$

Step 1: Split

$$x = z$$

$$\min_{x,z} \ \frac{1}{2}x^T Qx + q^T x + I(z)$$

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Step 2: ADMM Updates

x-update (key step)

$$x^{k+1} = \arg\min_{x} \frac{1}{2} x^T Q x + q^T x + \frac{\rho}{2} \| x - z^k + u^k \|^2$$

This gives linear system:

$$(Q + \rho I) x^{k+1} = \rho\left(z^k - u^k\right) - q$$

z-update (projection step)

$$z^{k+1} = \arg\min_{z} \left(I_{z \geq 0}(z) + \frac{\rho}{2} \| x^{k+1} - z + u^k \|^2 \right)$$

dual update

$$u^{k+1} = u^k + \left(x^{k+1} - z^{k+1}\right)$$

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